Not Every Concept Is a Direction:
Which Lexicalized Concepts Can Be Steered in LLMs? *

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Draft of July 3, 2026

Abstract

Steering vectors are widely used to control and interpret LLMs, yet benchmarks show that some concepts steer reliably while others barely respond — a variance that is reported but rarely explained. In this research we investigate which kinds of concepts are steerable in Llama-3.1-8B and Gemma-2-9B. We use the RFM method to extract steering vectors (or directions) for approximately 400 concepts drawn from 24 WordNet supersense categories, and validate which of them successfully steer the model. We then build a regression model explaining steerability from concept category, concreteness, polysemy, and word frequency, as well as the share of the top eigenvector during extraction (λ_1/trace). The results help distinguish concepts that behave as single directions — "primitive" for the model — from those that are composite or context-dependent, and offer practitioners an extraction-time indication of when single-vector steering can be trusted.

- Candidate sentence once results are in (pilot expectation): concreteness and polysemy — *not* frequency — predict steerability; the frequency null is itself a finding.
- Candidate sentence (pilot mechanism): when steering fails, it fails informatively — single-vector steering delivers the concept's *category prototype* (a city steers to "a beloved place", not to itself), because the extracted direction is dominated by what all category members share.
- Candidate sentence (causal probe): re-extracting with within-category contrasts tests whether such failures lie in the extraction design rather than the model.
- Candidate sentence (replication): findings replicate across two model families (Llama, Gemma).

1 Introduction

- Hook: steering/RePE papers throw heterogeneous concept lists at models; some concepts steer at 90%, others at chance. The variance is treated as noise; we argue it is signal.
- The implicit assumption of top-1 eigenvector / PCA-1 / difference-of-means steering: the concept is representable as *one direction*, injectable with *one coefficient*, at whatever layer(s) the method uses.
- Existing evidence that not all features are one-dimensional (Engels et al.) — but no one has asked *which* concepts behave as single directions, or whether this aligns with human semantic structure.

*Working title. Alternative: "Concreteness and Polysemy Predict the Steerability of Lexicalized Concepts" (use if R2 confirms).

- Our question: is steerability systematic and predictable from the linguistic properties of the word — and what does a steering failure actually consist of?
- Contributions:
  - C1: a systematic steerability study — ∼400 lexicalized concepts across 24 WordNet supersenses, annotated with psycholinguistic norms, steered under a controlled two-configuration protocol; per-concept results and code released.
  - C2: a regression explaining steerability variance from concept category, concreteness, polysemy, and frequency, with extraction-internal spectral metrics (λ_1/trace, λ_1/λ_2) as candidate zero-cost predictors.
  - C3: a mechanistic account of what failure looks like, grounded in pilot diagnostics (parent-dominance of extracted directions; coherence $\neq$ identity) and tested with one causal probe (within-category contrast rescue).

2 Background and Related Work

- **Linear representation hypothesis and steering.** Park et al.; RepE (Zou et al.); CAA (Panickssery/Rimsky et al.); ActAdd (Turner et al.); ITI (Li et al.); the RFM/universal-steering benchmark we build on (Beaglehole et al. 2026) — recap: kernel ridge + AGOP metric learning, top eigenvector as steering vector.
- **Multi-dimensional features.** Engels et al. (not all features are one-dimensional; circular features); feature splitting in SAEs; conceptors / subspace steering — subspace methods beat single vectors precisely where concepts are multi-dimensional.
- **Predicting steerability.** LAP / logit-lens accessibility (Billa 2026); "When is your LLM steerable?"; (un)reliability of steering vectors. Our wedge: these are engineering predictors; none ask about the linguistic structure of which concepts steer.
- **Causal identification of concept vectors.** Function vectors (Todd et al. 2024); linear relation embeddings (Hernandez et al. 2024); ROME (Meng et al.) — identification by *intervention*, complementary to our identification by *contrast design*.
- **Frequency and linear representations.** Co-occurrence thresholds for linear representations; long-tail knowledge (Kandpal et al.). Sets up the frequency hypothesis we then test (and, per pilot, expect to reject).
- **Psycholinguistic norms in NLP.** Brief: concreteness/AoA norms used for grounding and interpretability probes.
- Gap statement: no work connects concept-level semantic structure (concreteness, polysemy, taxonomic class) to steering outcome, nor explains what an unsteerable concept's failure consists of.

3 Defining and Sampling Concepts

- **Operational definition.** Sidestep the philosophical debate (Murphy 2002; Margolis & Laurence 1999): a *lexicalized concept* = a meaning a speech community assigned a word to, operationalized as a WordNet synset; each word mapped to its dominant (first) noun sense.

- **Inventory.** All single-word, noun-dominant lemmas from WordNet; category = supersense (26 lexicographer files; Miller 1990; Fellbaum 1998; "supersense": Ciaramita & Johnson 2003).
- Exclusions: `noun.Tops` (steering toward "entity" is ill-posed); POS-ambiguous lemmas filtered via dominant-POS + presence in norm lists; small cells (`motive`, `relation`) sampled exhaustively and flagged.
- **Sample (≈400, tiered).** Base 15/supersense across ∼24 usable supersenses; oversampled to ≈24 in six *featured* supersenses pre-specified before extraction: `location`, `person`, `feeling` (pilot continuity) and `artifact`, `cognition`, `food` (concreteness extremes). Featured cells carry the within-category analyses and absorb attrition; non-featured cells serve the pooled regressions, where per-cell n is not load-bearing. Within each supersense, stratify over frequency so category and frequency are not confounded.
- **Predictors** (one table): Zipf frequency (wordfreq); concreteness (Brysbaert et al. 2014); age of acquisition (Kuperman et al. 2012); polysemy (#WordNet senses); taxonomic depth (covariate); supersense; log category size (covariate).
- **Contrast design.** Positives wrap a statement in a persona frame; the frame must be felicitous across *all* supersenses (e.g. "You are fascinated by {concept}"), else frame felicity confounds category. *Dual negatives*: (i) bare statement (benchmark-comparable); (ii) same frame with a *different* concept — free, since those prompts are other concepts' positives — which cancels the frame axis, and with same-supersense negatives also the category parent. Rationale: a contrast in which frame, category, and instance always co-occur cannot statistically separate them (see Pilot).
- **Pilot/confirmatory split.** Pilot = 90 concepts (30 each: places, moods, personalities) from the released benchmark lists, used for all exploration; the WordNet sample is confirmatory, untouched during exploration.

4 Method

- **Direction extraction (RFM) — and why RFM.** Per layer: kernel ridge regression on concept-vs-negative prompts; AGOP $M = \mathbb{E}[\nabla f \nabla f^{\top}]$; steering vector = top eigenvector of M. Chosen among RepE methods because (a) it is the method of the benchmark we build on (comparability); (b) unlike difference-of-means or PCA-1 it returns the *full discriminative spectrum* with eigenvalues — the spectral metrics below come for free, and the method is self-diagnosing (the pilot's confound was discovered through them); (c) its steering performance matches or beats the simpler extractors on the released benchmark.
- **Spectral metrics.** $\lambda_1/\mathrm{tr}(M)$ (share of the spectrum on the top direction) and λ_1/λ_2; per-concept score = band average over decoder depth 4–24 (per-layer profile retained). Interpretation deferred to Pilot: these measure the *coherence* of the extracted signature, not concept identity.
- **Fixed hyperparameters — a methodological requirement.** Kernel bandwidth inflates λ_1/trace mechanically (large bandwidth ⇒ near-linear fit ⇒ rank-1 AGOP); grid search over near-tied validation correlations injects artifact variance into the dependent measure. All extractions at bw=1, norm=True, reg=10^{-3}; both vector and metrics read at a fixed (converged) iteration.
- **Two-config steering protocol.** Each concept steered under (i) *all layers*, benchmark defaults (unit vectors, released coefficient grid — comparability); (ii) *one mid layer* (≈40% depth; vector

scaled by the concept's per-layer projection gap, cf. CAA mean-difference magnitude / ITI σ-scaling; effective single-site dose $\approx 1\times$ local residual-stream norm). $\Delta(\text{all} - \text{single})$ measures how *distributed* the concept's steerable content is. 3 category-neutral prompts (listed verbatim); greedy decoding.

- **Outcome.** Primary: fraction of (coef × prompt) cells judged as expressing the concept; explicitly *not* max-over-coefficients. Secondary: *granularity reached* (category / region / instance) — the binary criterion flattens near-misses (pilot: Edinburgh→"Scottish Highlands" scores 0), and granularity is the variable the mechanism moves.
- **Judge.** Claude Haiku 4.5 via Batch API; symmetric judge question across supersenses; validation: dual-judge agreement (vs. GPT-OSS) on n=200 + human spot-check (κ reported).
- **Controls.** Prompted ceiling (can the model express the concept when explicitly asked?) — steerability conditional on ceiling; pipeline sanity-checked on known-good concepts per model.
- **Models.** Full protocol on Llama-3.1-8B-Instruct; headline analyses replicated on Gemma-2-9B-it.

5 Pilot Diagnostics and Protocol Selection

- **Hyper-parameter selection (summary).** Validation r saturates ($\sim$0.97–0.99) across bandwidths, statement counts, and iterations $\Rightarrow$ selection by validation is third-decimal noise; large bandwidth forces rank-1 AGOP by construction; converged readout cuts metric instability 3–4×. Decision rule: fixed modal setting, converged readout, bandwidth per model; details and sweep factors in Appendix A.
- **Finding 1: parent-dominance.** Different same-category concepts' extracted top-1 directions are nearly identical (places: cosine 0.97 → 0.84 → 0.72 across depth; *all* top-3 components $\geq$0.90 shared). Single-layer dose-response saturates at the category *prototype* (every city → "quiet beach at sunset") with region-level near-misses (Edinburgh→Dublin pub; genomicist→neuroscientist); giving secondary components injection mass adds a generic role-play axis, not instance content. Three independent legs (geometry, dose-response, component ablation); one figure: `parent_share_by_depth.png`. Interpretation: what single-vector steering delivers is mostly the category, because a contrast in which frame, parent, and instance always co-occur is statistically one axis.
- **Finding 2: coherence $\neq$ identity.** Places show the *highest* band λ_1/trace yet fail single-layer steering; moods (lower) succeed. Spectral metrics certify that a *coherent* (context-invariant, rank-1) direction was extracted — not that it is the judged content. Consequence: their regression coefficients are expected to be conditional on contrast type; tested under both contrasts.
- **Finding 3: attribute vs. instance.** Attribute concepts (moods) steer from one mid layer; instance identities steer only under all-layer injection, reaching region granularity — consistent with multi-stage assembly (per-layer instance-residue accumulation + late lexical tipping + autoregressive bootstrap once the name is emitted). Motivates the two-config protocol and the Δ measure.
- **Protocol justifications** (one line each; evidence in Appendices C–D): single mid layer at $\approx$40% depth — wide behavioral window there; 60% depth yields token-bias then repetition collapse. No top-k arms — the top-3 subspace under the released contrast is category-shared, so k cannot add

instance content. Coefficient $\approx 1\times$ local residual norm via per-layer magn scaling. Granularity as secondary outcome — the binary judge flattens near-misses.

6 Results

- **R1: Steerability varies systematically.** Steering success varies widely across concepts and clusters by supersense; distribution figure per supersense, per config.
- **R2: Psycholinguistic predictors.** Regression of steerability on concreteness, polysemy, AoA, Zipf, taxonomic depth (+ category size). Pilot expectation: concreteness and polysemy carry the signal; frequency ≈ 0 — exposure alone does not buy steerability.
- **R3: Spectral metrics as candidate predictors.** λ_1/trace and λ_1/λ_2 within the full regression; validity compared across the two contrast types (expected conditional — see Pilot Finding 2); controlling for prompted ceiling and val-r (saturated $\Rightarrow$ cannot explain variance). Baselines: frequency alone, category alone.
- **R4: The rescue probe (causal).** For instance-like concepts that fail: re-extract with within-category contrasts ("loves Edinburgh" vs. "loves {other city}"; frame and parent cancel) and steer at one mid layer. Success $\Rightarrow$ the failure was extraction addressing (contrast design), not model capability; failure $\Rightarrow$ instance content is not single-site injectable (assembly account).
- **R5: Distributedness.** Δ(all-layer − single-layer) per concept, reported descriptively by supersense and correlated with the R2 predictors.
- **R6: Generality.** Llama vs. Gemma consistency: per-concept correlation of steerability and of the spectral metrics across models.

7 Discussion

- What makes a concept steerable: attributes (context-general colorings) admit a single injectable direction; instance identities require assembled, multi-stage content — steering failure is usually *delivery of the parent category*, not silence.
- What λ_1/trace is: an *intervention-match score* — how well the concept's natural activation signature fits the one move single-vector steering can make (one constant vector, all contexts). Mechanism for R2: concrete concepts express context-invariantly (rank-1 signature); abstract ones express context-dependently.
- Status of extracted vectors: the residual stream has no privileged basis (rotations absorb into reader/writer weights), so extracted directions are *experiment-relative*; within-model relational claims (cosines, subspaces) are valid, coordinate-level and raw cross-model comparisons are not. Anchored spaces — embedding/unembedding (Park's causal inner product), MLP-neuron space (Geva) — are where mechanism pins a frame.
- Two routes to a vector's identity: *experiment design* (contrasts that vary exactly one thing) vs. *causal mediation* (function vectors, ROME). Pipeline: propose by contrast, accept by causal test — R4 instantiates it.
- Practical implications for RePE: report spectral concentration and contrast design alongside steering benchmarks; concept inventories should be stratified, not ad hoc; granularity-graded judging over binary.

Future work: abstraction-depth as a second axis

- Idea: besides *how many* directions compose a concept, *where* (which layer) it first becomes steerable — hypothesis: hypernyms steer shallower than hyponyms (cf. pilot: moods at depth $\sim$9, places at $\sim$17+).
- Pilot lesson: this is a *write* property — extraction-geometry proxies (direction rotation, alignment-to-deep) are concept-invariant and fail the steering ground truth; measurement requires single-layer steering sweeps.
- Test on the WordNet hierarchy (taxonomic depth as ground truth); independence from the compositional axis must be assessed *within* supersense (between classes the two are confounded).
- Related vignette: compositional concepts — is direction("strict father") $\approx \alpha$ direction("strict") $+\beta$ direction("father")?

8 Limitations

- Single extraction method (RFM); λ_1/trace is AGOP-specific — PCA explained-variance as cross-check (appendix if time).
- Steering conclusions are relative to the additive intervention class; clamp/projection interventions may behave differently (targets saved, untested).
- LLM judge (validated but imperfect); English only; dominant-sense assumption; norms are human ratings.
- Correlational core with one causal probe (R4); extraction-design effects and model limits not fully separable outside that design.

A Appendix A: Extraction and metric setup

+ Validation r (probe accuracy) saturates ($\sim$0.99) across bandwidths, statement counts, and RFM iterations $\Rightarrow$ any selection *by* val-r is third-decimal noise; it is also a poor steerability predictor. Boxplots: `box_rsq.png`.
+ λ_1/trace at bandwidth L=1 discriminates across concepts; L=10 collapses it toward 1 (rank-1 AGOP by construction). `box_l1trace.png`, `box_l1l2.png`.
+ The released λ_1/trace (read at the best-r iteration) is unstable — it samples a value climbing 0.0002 $\rightarrow$ 0.17 $\rightarrow$ 0.46 across iterations at a noise-chosen point; reading at convergence cuts the statement-count swing 3–4$\times$. Both vector and metrics read at a fixed (converged) iteration.
+ Saved per-layer extraction stats: top-3 eigenvalues (`evals`), natural concept magnitude (`magn` = raw projection gap), clamp targets (`tpos`, `tneg`), mean residual norm (`hnorm`) — in the rfmstats pickles.

? Per-concept "best layer" by argmax is unstable: val-r best is noise-driven; λ_1/trace and λ_1/λ_2 are bimodal with a degenerate spike at the first block (exclude early blocks). `besthist_*.png`.

? Hyper-parameter sweep factors (probe set of 24 concepts, one per supersense): bandwidth $\times$ normalization $\times$ contrastive construction (now incl. within-category negatives) $\times$ iteration read-out; criteria: stability, discrimination, semantic signal (Spearman vs. concreteness/polysemy), steering non-degradation. Artifacts: `hpsweep_phase2_*`.

B Appendix B: Predicting steerability on the pilot (Llama-3.1-8B, 3 classes)

+ Best single predictor of steering success is λ_1/trace at a mid layer (depth 9, grouped CV-AUC ≈0.86); λ_1/λ_2 similar; val-r erratic (saturated). Logistic regressions in `nb_regxp1.ipynb`; `L9_metrics_by_steerability.png`.
– Multivariate regression on *all* layers' metrics is uninformative — severe multicollinearity (the residual stream is cumulative). Univariate per-layer CV-AUC is the clean read.
+ **Sign flips with depth**: λ_1/trace predicts steerability *positively* early (depth 9) but *negatively* late (depth 29) — late-layer concentration is lexical collapse, not concept concentration. `l1l2_depth29_dist.png`.
? Concept class is a confounder/mediator: most steerability variance is between-class; within-class signal is weak. Do not "control for class" if class is on the causal path (concreteness → signature coherence → steerability).
? **The inversion**: places have the *highest* band-averaged λ_1/trace yet fail single-layer steering, while moods (lowest) succeed — the metric certifies *coherence* of the extracted blend, not its *identity*. Predictor validity is contrast-dependent (main text, Pilot Finding 2).
? Participation ratio ≈2–3 for all concepts at all layers — the discriminative subspace at one layer is effectively rank 2–3, bounding what top-k can recover. `PR_depth9_hist.png`.
+ Error analysis at depth 9: misclassifications symmetric, in a narrow overlap band; characterizes the ∼0.86 ceiling.

C Appendix C: Steering interventions (what moved the needle)

– Single-layer steering at depth 9: moods steer, experts middling, places ≈0 — places steer to a *generic* place attractor, not the named city (judge scores correct, not an artifact).
+ Two mid layers (9+17) was the best targeted multi-site result (∼48/35 of 90); 9+26 worse (late layer contributes generic content); late-only steering fails outright.
+ **Cross-layer directions are near-orthogonal** (cosine ≈0.07): the concept is re-encoded per layer, so two layers are complementary slices while extra components at one layer are not.
– Single-site depth-19: *no behavioral window* — default → lexical word-lists (bitter → "bitter, cold, dry, sarcastic") → repetition collapse within 0.5–2× local stream norm; moods 10/15 (partly lexical credit), places/personalities ≈0. Late injection biases output *tokens*, not behavior.
+ Single-site depth-13: *wide* behavioral window (embodied persona, clean saturating dose-response, no collapse): moods 15/15; places 0/15, saturating at the category prototype ("quiet beach at sunset" for every city) with region-level near-misses (Edinburgh→Dublin pub; genomicist→neuroscientist). Coefficient is not the limiting factor; *vector content* is.
– Top-k at one layer, k=3: with eigenvalue weights ≈ k=1; with equal weights *worse* (moods 15→10) — components 2–3 are a generic roleplay/self-narration axis ("my name is Luna"), 0.92/0.90 shared across cities. **Parent-dominance**: all top-3 components ≥0.90 shared across same-category concepts at every depth (places cosine 0.97 → 0.84 → 0.72; moods individuate to 0.46) — the contrast, in which frame, parent, and instance always co-occur, is statistically one axis. `plots/parent_share_by_depth.png`.

+ All-layer steering (benchmark defaults): 26/45 v1; places reach *region* granularity (Edinburgh→Scottish Highlands with Scots dialect; Austin named exactly) — per-layer instance-residue accumulation + late lexical tipping + autoregressive bootstrap (once the name is emitted, context sustains it).

? Sample note: the depth-13/19, k=3, and all-layer bullets above report the interim n=15/class shakedown; all four arms to be re-run on the standard 90-concept pilot (extraction in `direction3e`; generation+judging only). Qualitative patterns expected to hold.

? **Planned probe — within-category contrast extraction** (= R4): positives "loves Edinburgh", negatives "loves {other city}" (same frame and statements both sides) — frame and parent cancel; the only label-varying signal *is* the city. Steer single-layer (depth 13, magn-calibrated) on ∼5 cities. Cities named ⇒ failure was extraction *addressing*; still generic ⇒ instance content is not single-site injectable (assembly account).

? Zero-reextraction pilot of the probe: $v_{\text{spec}} = v_{\text{city}} - \overline{v}_{\text{other cities}}$ from saved per-layer directions (the shared component cancels in the difference); noisier but free — run before committing extraction compute.

D Appendix D: Coefficient calibration (systematic magnitude)

– Fixed/hand-swept coefficients are brittle: the working value shifts ∼10× with depth and with the number of steered layers.

+ `magn` (natural projection gap, saved per layer) grows ∼exponentially with depth; scaling each layer's vector by `magn` makes the coefficient a dimensionless multiplier (cf. CAA mean-difference magnitude; ITI σ-scaling). `magn_by_layer.png`.

+ $\sqrt{N}$ count-normalization (cross-layer directions ≈orthogonal ⇒ quadrature) makes the multiplier roughly transfer across layer counts; usable band centers near ∼1.

+ Single-site calibration: effective single-layer steering needs ≈1.0× the local residual norm (`hnorm`); `magn/hnorm` ≈ 0.4–0.5 mid-stream ⇒ multiplier ≈2–3. Empirically `magn` ≈ norm-relative scaling (r=0.90 with `hnorm`), so the two schemes are redundant.

? Clamp / project-to-target (set the projection to the concept's natural positive-class value; `tpos/tneg` saved) is the one structurally different alternative — bounded, token-dependent, self-allocating across layers; untested.

? Process lesson: three config-state incidents (stale judge CSVs, judge-cache pickles, stale coefficient list) share one root cause — hand-set knobs without provenance. Fix: auto-derive coefficient grids from saved per-layer stats; encode a run tag in artifact filenames.

E Appendix Z1 (parked): Localizing the concept — "where is Paris?" (diagnostics)

+ **ROME causal trace** ("Paris is the capital of"→France): the fact lives at the *subject* token in early–mid layers (∼0–16) and is relayed to the final token only late (∼14–31). `ROME-PARIS.png`; `diag_paris_rome.ipynb`; `diag_paris_rome_ha.ipynb`.

+ This explains the RFM mismatch: RFM reads the *last* token, where subject-specific content arrives only late $\Rightarrow$ early extraction captures an *abstraction*, late the *precise* instance.

+ **SAE features** confirm the model holds Paris: clean Paris / France / city features at the steered layers (Goodfire l19; Llama-Scope L9/17/19, Neuronpedia labels). `diag_paris_sae.ipynb`, `diag_paris_sae_2.ipynb`. The steering failure is extraction/geometry, not absence.

? **Tegmark spatial probe** (lat/lon linear probe): coordinates linearly decodable but underpowered at N=90 cities (median $\sim$2200 km). `diag_paris_tegmark.ipynb`.

+ **Where "late" begins**: logit-lens KL-to-output gives stage boundaries to avoid the residual-sharpening tail. `diag_stages_late.ipynb`.

F Appendix Z2 (parked): Conceptual threads

? **Abstract-early / precise-late hierarchy**: the extraction layer sets the abstraction level; the precise concept lives late (weak steering) while the strong injection point is mid (concept abstract) — no single layer gives both.

? **Two candidate axes** (now Discussion future work): (i) abstraction-depth — first layer that can *steer* the concept; (ii) compositional basis size (λ_1/trace). Between-class confounded (moods both more composite *and* earlier-steering than places); independence is a within-supersense question.

– Extraction geometry cannot proxy first-steerable-depth: rotation and alignment-to-deep proxies are concept-invariant ($\sim$17–20) and fail the steering ground truth (cross-layer orthogonality, $\cos \approx 0.05$). A write property; measure by steering. `nb_axes.ipynb`; `plots/axes_independence.png`.

? **Superposition vs. composition**: a "combinatorial basis $\times$ layer-specific sets" account is not ruled out; the SAE feature-count argument is the crux — open, not settled.

? **Stages of inference / lens** framing (Lad–Gurnee–Tegmark) as the principled naming of early/mid/late bands.

? **Not-all-linear** (Engels et al., arXiv:2405.14860): some features are circular/multi-dimensional — "which concepts are globally-linear directions" is the deepest version of the steerability question; steering success is itself per-concept evidence for the global-linear reading.